Full Unimodular Cyclicity via a Bilateral Fixed-Space Dilation

Candidate full solution for the FA/Banach discovery run

23 July 2026

Source problem and answer

The source is Jochen Glück, *A note on the fixed space of positive contractions*, arXiv:2108.05174 [1]. Its Open Problem 3.4 on page 6 asks whether Corollary 3.3 remains true when the eigenvalue λ is an arbitrary unimodular number rather than a root of unity. Thus, for a positive contraction T on a monotonically complete complex Banach lattice with the Fatou property, the question asks whether

$$\dim\ker(\lambda - T) \le \dim\ker(\lambda^k - T) \qquad (k \in \mathbb{Z})$$

whenever $|\lambda| = 1$.

The answer is affirmative. The source statement and its immediate context are reproduced below.

> But for lattice homomorphisms, the claimed dimension estimate is known. More precisely, we have for each integer k
>
> $$\dim\ker(\lambda - T) = \dim\ker(\lambda - T|_F) \le \dim\ker(\lambda^k - T|_F) \le \dim\ker(\lambda^k - T),$$
>
> where the estimate in the middle is a property of lattice homomorphisms that can be found in [10, Proposition 3.1]. So the claimed dimension estimate holds; in particular, λ^k is an eigenvalue of T. $\square$
>
> The proof of Corollary 3.3 relies heavily on the assumption that λ be a root of unity. Thus, the following question naturally arises:
>
> **Open Problem 3.4.** Does the assertion of Corollary 3.3 remain true if λ is not a root of unity, but more generally a complex number of modulus $|\lambda| = 1$?

A bounded novelty search on 23 July 2026 used the arXiv id and title, the exact wording of Open Problem 3.4, and combinations of *positive contraction*, *monotonically complete*, *Fatou property*, *peripheral point spectrum*, *cyclicity*, and *bilateral shift*. Searches of the local 2000–2026 arXiv source corpus and web/arXiv-oriented results found the source paper, its 2016 predecessor [2], and bibliographic mirrors, but no later work claiming the result below. This is bounded negative evidence, not a comprehensive novelty guarantee.

Proof intuition

The root-of-unity proof in the source restricts T to $\mathrm{Fix}(T^n)$, where it becomes a lattice automorphism. For an irrational unimodular eigenvalue there is no such n. The replacement is to manufacture a fixed space on which an auxiliary operator is automatically invertible.

Pass to the bilateral sequence lattice $\widehat{E} = \ell^\infty(\mathbb{Z}; E)$ and set

$$(Sx)_n = Tx_{n-1}, \qquad (Rx)_n = x_{n+1}.$$

The operator S is a positive contraction, so its fixed space F is a Banach lattice by the source paper's main theorem. The bilateral shift R commutes with S; hence $R|_F$ is a lattice automorphism. An eigenvector $Tz = \lambda z$ is encoded by the fixed sequence $(\lambda^n z)_{n\in\mathbb{Z}}$, which is a λ-eigenvector of $R|_F$. Cyclicity for lattice automorphisms produces the λ^k eigenspaces, and evaluation at coordinate zero transfers them back to T.

Two elementary lemmas

Recall that a Banach lattice E has the Fatou property if $0 \le x_i \uparrow x$ implies $\|x\| = \sup_i \|x_i\|$, and it is monotonically complete if every increasing norm-bounded net in E_+ has a supremum.

Lemma 1 (stability under bounded products)**.** *If E is monotonically complete and has the Fatou property, then $\ell^\infty(I;E)$, with its coordinatewise order and supremum norm, has both properties for every index set I.*

Proof. Let (x^α) be an increasing norm-bounded net in $\ell^\infty(I;E)_+$, with $\|x^\alpha\| \le C$. For each $i \in I$, monotone completeness gives

$$x_i := \sup_\alpha x_i^\alpha \in E.$$

The Fatou property of E gives $\|x_i\| = \sup_\alpha \|x_i^\alpha\| \le C$, so $x = (x_i)_{i\in I}$ belongs to $\ell^\infty(I;E)$ and is plainly the supremum of the net. Finally,

$$\|x\| = \sup_{i\in I}\sup_\alpha \|x_i^\alpha\| = \sup_\alpha\sup_{i\in I} \|x_i^\alpha\| = \sup_\alpha \|x^\alpha\|,$$

which is the Fatou property. □

Lemma 2 (cyclicity for a lattice automorphism)**.** *Let V be a complex Banach lattice and let U be a lattice automorphism of V. For every $\mu \in \mathbb{T}$ and $k \in \mathbb{Z}$,*

$$\dim\ker(\mu - U) \le \dim\ker(\mu^k - U).$$

Proof. This is the standard lattice-power argument used in [2, Proposition 3.1] and invoked by the source in its proof of Corollary 3.3. We recall the dimension mechanism.

For $0 \ne f \in V$, represent the principal ideal generated by $|f|$ as a $C(K)$-space and define the lattice power $f^{[k]}$ pointwise by

$$f^{[k]}(t) = \begin{cases} \big(f(t)/|f(t)|\big)^k |f(t)|, & f(t) \ne 0,\\ 0, & f(t) = 0.\end{cases}$$

This definition is intrinsic. Since lattice homomorphisms preserve lattice powers,

$$f \in \ker(\mu - U)\setminus\{0\} \quad\Longrightarrow\quad f^{[k]} \in \ker(\mu^k - U).$$

For completeness, this implication yields the dimension inequality even though $f \mapsto f^{[k]}$ is nonlinear. Given m linearly independent vectors in $\ker(\mu - U)$, work in the principal ideal generated by the sum of their moduli. In its $C(K)$-representation, choose m points and a new basis $f_1, \dots, f_m$ of their span with $f_j(t_\ell) = \delta_{j\ell}$. Then $f_j^{[k]}(t_\ell) = \delta_{j\ell}$, including for $k \le 0$ under the zero convention above. Hence the $f_j^{[k]}$ are linearly independent in $\ker(\mu^k - U)$. □

Full solution of Open Problem 3.4

Theorem 3. *Let E be a complex Banach lattice which is monotonically complete and has the Fatou property, and let $T \in \mathcal{L}(E)$ be positive with $\|T\| \le 1$. If $\lambda \in \mathbb{T}$, then for every $k \in \mathbb{Z}$,*

$$\dim\ker(\lambda - T) \le \dim\ker(\lambda^k - T).$$

In particular, if λ is an eigenvalue of T, then every integer power λ^k is an eigenvalue of T.

Proof. Set

$$\widehat{E} := \ell^\infty(\mathbb{Z}; E).$$

By Lemma 1, this complex Banach lattice is monotonically complete and has the Fatou property. Define $S, R \in \mathcal{L}(\widehat{E})$ by

$$(Sx)_n := Tx_{n-1}, \qquad (Rx)_n := x_{n+1} \qquad (n \in \mathbb{Z}).$$

The operator S is a positive contraction, R is a lattice isometric automorphism, and $SR = RS$. Let

$$F := \operatorname{Fix} S.$$

Theorem 3.1 of the source [1] shows that F, with the inherited order and norm, is a monotonically complete complex Banach lattice with the Fatou property. Since both R and R^{-1} commute with S, they leave F invariant. Consequently $R|_F$ and its inverse are positive, so $R|_F$ is a lattice automorphism of F.

For each $\mu \in \mathbb{T}$, define

$$J_\mu : \ker(\mu - T) \longrightarrow F, \qquad (J_\mu z)_n := \mu^n z.$$

If $Tz = \mu z$, then

$$(SJ_\mu z)_n = T(\mu^{n-1} z) = \mu^n z, \qquad (RJ_\mu z)_n = \mu^{n+1} z = \mu (J_\mu z)_n.$$

Thus J_μ maps into $\ker(\mu - R|_F)$. Conversely, if $x \in F$ and $Rx = \mu x$, then $x_n = \mu^n x_0$, while $Sx = x$ at coordinate $n = 1$ gives

$$Tx_0 = x_1 = \mu x_0.$$

Hence evaluation at coordinate zero is the inverse of J_μ, and

$$\dim\ker(\mu - T) = \dim\ker(\mu - R|_F).$$

Apply Lemma 2 to $R|_F$:

$$\begin{aligned}\dim\ker(\lambda - T) &= \dim\ker(\lambda - R|_F)\\ &\le \dim\ker(\lambda^k - R|_F) = \dim\ker(\lambda^k - T).\end{aligned}$$

This proves the assertion for every $k \in \mathbb{Z}$. $\square$

Additional consequence: the semigroup dimension estimate

The same dilation also proves the dimension estimate anticipated immediately after Corollary 3.5 of the source.

Corollary 4. *Let E be as in Theorem 3, and let $(T_t)_{t\geq 0}$ be a positive contractive C_0-semigroup on E with generator A. Then, for $\beta \in \mathbb{R}$ and $k \in \mathbb{Z}$,*

$$\dim \ker(i\beta - A) \leq \dim \ker(ik\beta - A).$$

Proof. On $\widehat{E} = \ell^\infty(\mathbb{R}; E)$, define the commuting family of positive contractions

$$(S_t f)(r) := T_t f(r - t) \qquad (t \geq 0,\ r \in \mathbb{R}),$$

and let $F = \bigcap_{t\geq 0} \operatorname{Fix} S_t$. Lemma 1 and Theorem 3.1 of [1], now applied to the whole commuting family, make F a complex Banach lattice. Every translation $(R_u f)(r) = f(r + u)$, $u \in \mathbb{R}$, restricts to a lattice automorphism of F.

The map

$$z \longmapsto \left(r \mapsto e^{i\beta r} z\right)$$

is a linear bijection from $\ker(i\beta - A)$ onto

$$G_\beta := \{f \in F : R_u f = e^{i\beta u} f \text{ for every } u \in \mathbb{R}\}.$$

Indeed, the forward implication follows from $T_t z = e^{i\beta t} z$. Conversely, a member of G_β has the form $f(r) = e^{i\beta r} f(0)$, and $S_t f = f$ at $r = t$ yields $T_t f(0) = e^{i\beta t} f(0)$, which is equivalent to $f(0) \in \ker(i\beta - A)$.

If $0 \neq f \in G_\beta$, then every R_u is a lattice automorphism, so

$$R_u(f^{[k]}) = (R_u f)^{[k]} = e^{ik\beta u} f^{[k]}.$$

Thus $f^{[k]} \in G_{k\beta}$. The finite-dimensional separation argument in Lemma 2, applied to G_β and $G_{k\beta}$, gives $\dim G_\beta \leq \dim G_{k\beta}$, which is the claimed estimate. □

Verification notes and limitations

- The bilateral construction uses all bounded E-valued functions. There is no measurability, separability, or strong-continuity assumption hidden in Lemma 1.
- Both shifts are genuinely bilateral. This is what makes $R|_F$ invertible with a positive inverse.
- The sign convention was checked in both directions: $S(\mu^n z)_n = (\mu^n z)_n$, and $Sx = x$, $Rx = \mu x$ imply $Tx_0 = x_1 = \mu x_0$.
- The only non-elementary input is Theorem 3.1 of the source, together with the standard lattice-power dimension lemma already used by the source in Corollary 3.3.
- The result does not extend merely by replacing contractivity with power boundedness: the source gives a counterexample under that weaker hypothesis.
- The bounded literature search cannot exclude an unindexed or differently phrased proof. Novelty confidence is moderate.

References

[1] J. Glück, *A note on the fixed space of positive contractions*, arXiv:2108.05174, 2021.

[2] J. Glück, *On the peripheral spectrum of positive operators*, Positivity **20** (2016), 307–336; arXiv:1503.02430.